

ENVS 193DS Homework #3

Ella Menin

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Setting up

```
# reading in package: tidyverse
library(tidyverse)
# reading in package: here
library(here)
# reading in package: janitor
library(janitor)
# reading in package: readxl
library(readxl)
```

```
# storing read kelp data as "kelp" object
kelp <- read.csv(here("data", "temp-kelp.csv"))
```

```
# storing read personal data as "personal_data" object
personal_data <- read.csv(here("data", "personal data(Sheet1).csv"))
```

Problem 1: Giant kelp fronds

a. Appropriate tests

The appropriate tests to determine the strength of the relationship between ocean temperature and giant kelp frond elongation rate are a Pearson correlation and a Spearman rank correlation. Pearson correlation measures the strength and direction of a linear relationship between two variables (in this case, temperature and giant kelp frond elongation rate), while Spearman correlation measures the strength and direction of a relationship using ranked data rather than actual raw data values. A Spearman rank correlation is appropriate when assumptions of linearity or normality are not met.

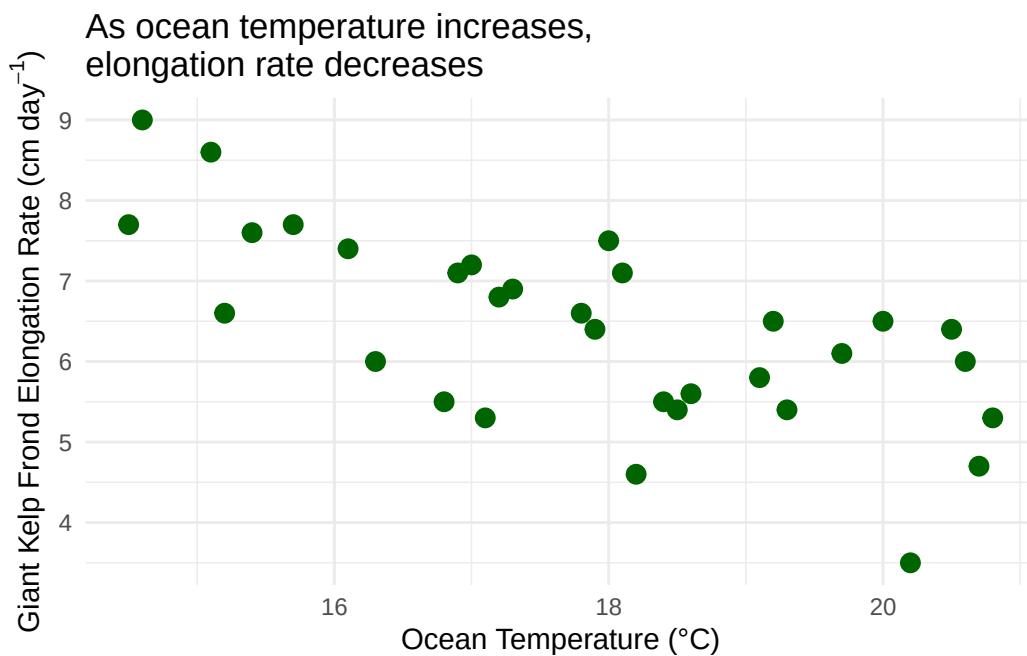
b. Creating a visualization

```
# base layer ggplot with kelp_preds data frame
ggplot(data = kelp,
       # setting aesthetics
       mapping = aes(
         # x is temperature measured in celsius
```

```

    x = temp_c,
    # y is giant kelp frond elongation rate measured in cm per day
    y = kelp_elong)) +
# first layer geom point showing individual data points
geom_point(color = "darkgreen",
           # making points noticeably large
           size = 3) +
# using labs to set axis titles
labs(
  # x axis: ocean temperature
  x = "Ocean Temperature (°C)",
  # y axis: giant kelp frond elongation rate
  y = expression("Giant Kelp Frond Elongation Rate (" * cm ~ day^{-1} *
    ↪ ")"),
  # visualization title: finding from visualization
  title = "As ocean temperature increases, \nelongation rate decreases"
) +
# setting a nice, basic theme
theme_minimal()

```



c. Checking assumptions and running a test

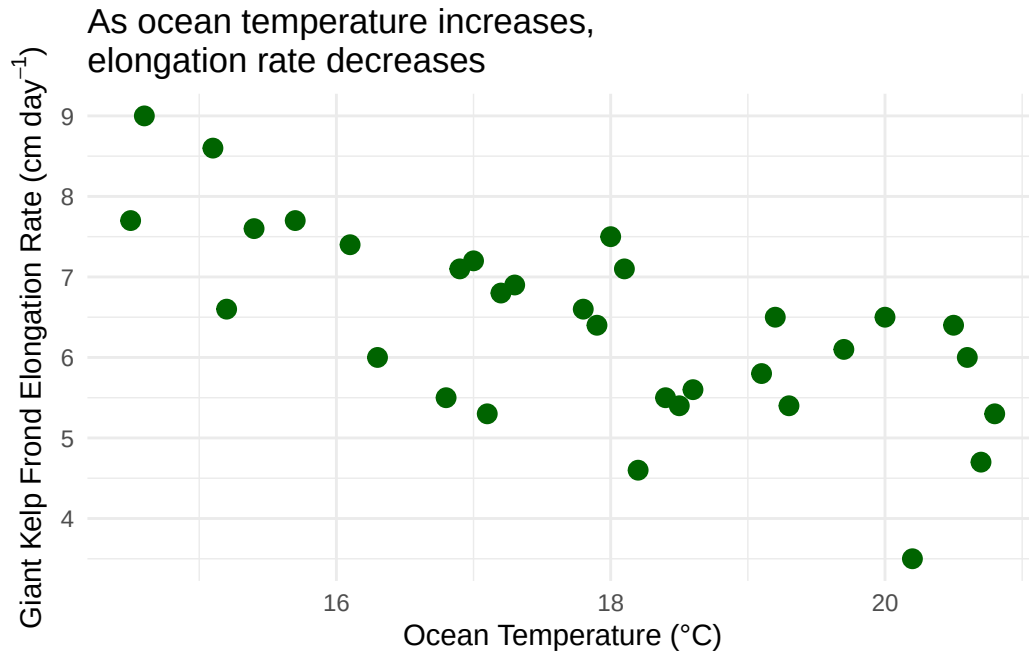
Checking assumptions:

To check the assumptions for a Spearman rank correlation, we must test linearity and the normality of the two variables: temperature and giant kelp frond elongation rate.

To test for linearity, we are going to utilize the previously made correlation visualization (shown again for viewer convenience).

```
# using previously made visualization to check for linearity

# base layer ggplot with kelp_preds data frame
ggplot(data = kelp,
       # setting aesthetics
       mapping = aes(
         # x is temperature measured in celsius
         x = temp_c,
         # y is giant kelp frond elongation rate measured in cm per day
         y = kelp_elong)) +
# first layer geom point showing individual data points
geom_point(color = "darkgreen",
          # making points noticeably large
          size = 3) +
# using labs to set axis titles
labs(
  # x axis: ocean temperature
  x = "Ocean Temperature (°C)",
  # y axis: giant kelp frond elongation rate
  y = expression("Giant Kelp Frond Elongation Rate (" * cm ~ day^{-1} *
    ↵ ")"),
  # visualization title: finding from visualization
  title = "As ocean temperature increases, \nelongation rate decreases"
) +
# setting a nice, basic theme
theme_minimal()
```

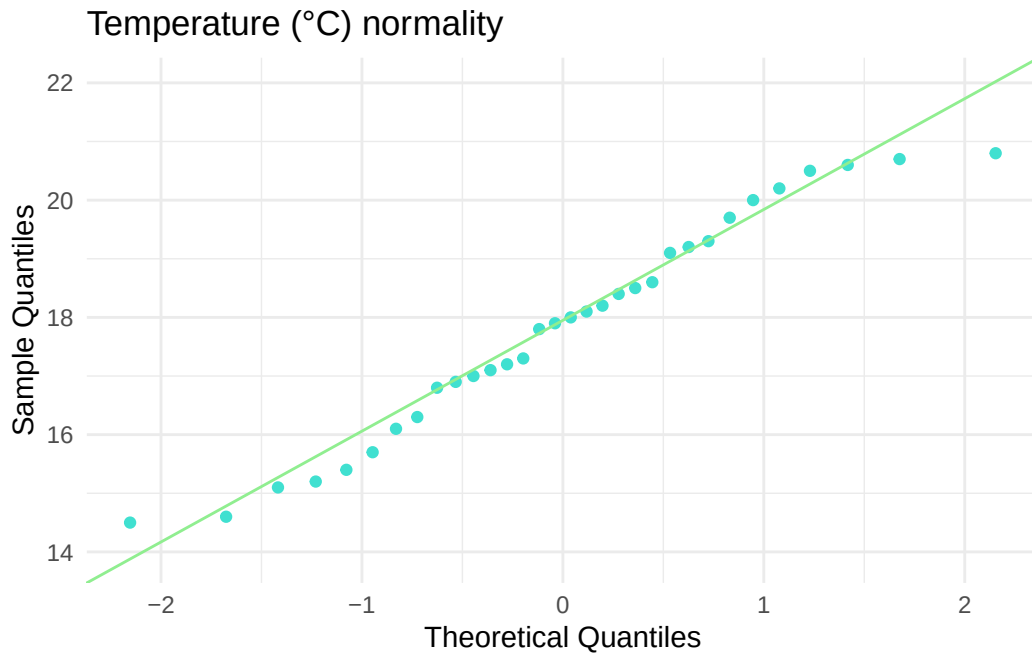


Based on the orientation of the points, temperature and giant kelp frond elongation rate have a negative linear relationship.

In order to test the normality of both variables, it is necessary to construct a QQ plot for each one.

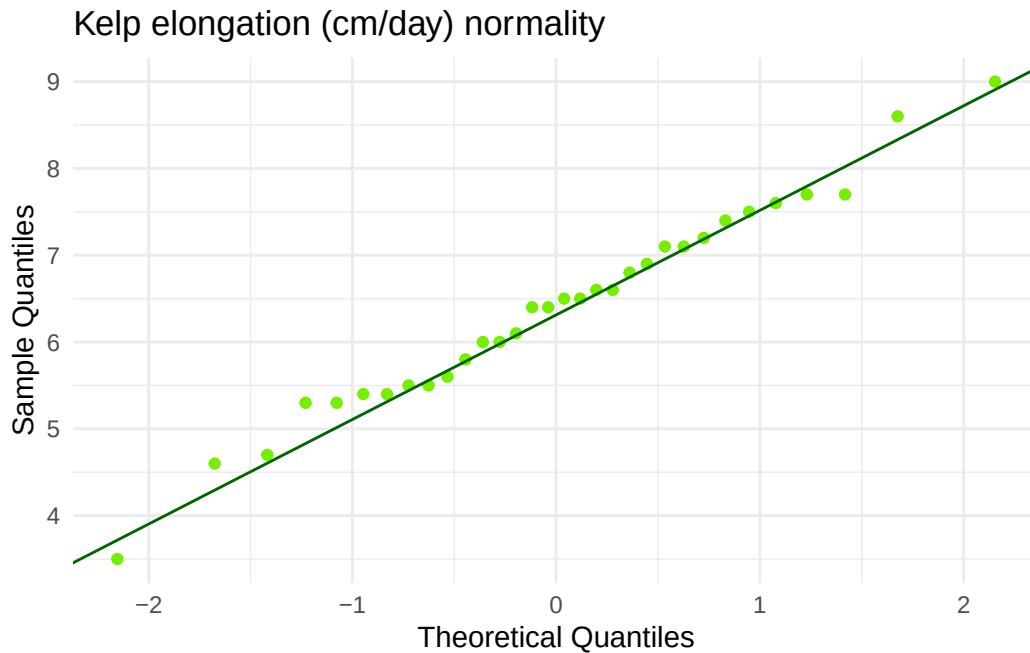
QQ plot for temperature data normality:

```
# base layer ggplot with kelp data frame
ggplot(data = kelp,
  # sample temp_c variable because we are testing normality of
  ↪ temperature
  mapping = aes(sample = temp_c)) +
# first layer individual data points in pretty turquoise color
geom_qq(col = "turquoise") +
# second layer is normality line in light green
geom_qq_line(col = "lightgreen") +
# relabeling axes
labs(x = "Theoretical Quantiles", # x axis theoretical quantiles
  y = "Sample Quantiles", # y axis sample quantiles
  title = "Temperature (\u00B0C) normality") + #title is temperature
  ↪ normality
# nice, minimal theme for aesthetics
theme_minimal()
```



QQ plot for kelp elongation data normality:

```
# base layer ggplot with kelp data frame
ggplot(data = kelp,
  # sample is kelp elongation values because we are checking normality
  ↪ of kelp elongation
  mapping = aes(sample = kelp_elong)) +
# first layer geom_qq shows individual data points in chartreuse
geom_qq(col = "chartreuse2") +
# second layer geom qq line shows normality line in dark green
geom_qq_line(col = "darkgreen") +
# labeling axes and title
labs(x = "Theoretical Quantiles", # axis is theoretical quantiles
  y = "Sample Quantiles", # y is sample quantiles
  title = "Kelp elongation (cm/day) normality") + # title conveys that
  ↪ we are testing kelp elongation normality
# nice, simple theme for aesthetics
theme_minimal()
```



Based on the QQ plots above, the sample data points roughly align with the the reference line, so we can determine that both variables are normally distributed enough.

Running tests:

```
# creating new object spearman_test from spearman rank test results
spearman_test <- cor.test(
  # first variable ocean temperature
  kelp$temp_c,
  # second variable giant kelp frond elongation rate
  kelp$kelp_elong,
  # running a spearman rank test to show correlation
  method = "spearman"
)

# display results
spearman_test
```

Spearman's rank correlation rho

data: kelp\$temp_c and kelp\$kelp_elong

```
S = 9216.1, p-value = 1.29e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
-0.6891684
```

d. Communicating results

To evaluate the strength of the relationship between temperature and giant kelp frond elongation rate, I used a Spearman rank correlation test because it can be used to measure the strength and direction of the correlation between two variables. After running this test, it has been concluded there is a moderate negative relationship between ocean temperature and giant kelp frond elongation rate. (Spearman rank correlation test, $\rho = 0.68$, $S = 9216.1$, $p < 0.001$, $\alpha = 0.05$).

Furthermore, according to our statistical test, ocean temperature does have an effect on giant kelp frond elongation rate, potentially because giant kelp may biologically thrive in particular ocean temperatures and conditions. This test suggests that they grow faster in colder temperatures.

e. Test implications

These results suggest that warmer ocean temperature ($^{\circ}\text{C}$) could reduce giant kelp frond elongation rate (cm day^{-1}). Because giant kelp forests provide crucial habitat and food resources for marine animals, a slower growth rate of these giant kelp fronds could result in resource availability for the dependent species. As the ongoing climate crisis worsens, ocean temperatures are projected to continue rising, which can further encourage the slowing of giant kelp frond elongation. Furthermore, it is crucial to continue investing time and funding into developing solutions to rising sea temperatures.

f. Double checking using a different test

```
# pearson correlation test to assess strength and direction of relationship
cor.test(
  # first variable: temperature from kelp data frame
  kelp$temp_c,
  # second variable: elongation rate from kelp data frame
  kelp$kelp_elong,
  # specification that it is a pearson correlation test
  method = "pearson"
```



```
)
```

Pearson's product-moment correlation

```
data: kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
      cor
-0.6877489
```

The Pearson correlation test and the Spearman rank correlation test produced the same overall conclusion: to reject the null hypothesis in favor of accepting the alternative hypothesis, as both tests found a statistically significant negative relationship between ocean temperature and giant kelp frond elongation rate.

The Pearson correlation measured the strength and direction of the relationship between temperature and kelp elongation rate ($r = -0.68$), and so did the Spearman test, yielding an almost exactly the same correlation coefficient ($\rho = -0.69$).

Problem 2: Personal data

a. Updating my visualizations

Figure 1:

```
# creating a new object specifically for my cleaned data
my_data_clean <- personal_data |>
# cleaning up the names to include lowercase letters and underscores
janitor::clean_names() |>
filter(!is.na(temperature_h_w_c),
       temperature_h_w_c != "") |>
mutate(temperature_h_w_c = factor(temperature_h_w_c,
                                  levels = c("H", "W", "C"), # order in which
                                   ↪ levels will appear on plot
                                  labels = c("Hot", "Warm", "Cold"))) # full
                                   ↪ words for plot labels
```

```

# base layer ggplot with my_data_clean data frame
ggplot(data = my_data_clean,
      # setting x and y axes
      mapping = aes(x = temperature_h_w_c, # x axis temperature
                    y = water_intake_l, # y axis water intake in liters
                    color = temperature_h_w_c)) + # color by temperature
# first layer geom_boxplot to assess median amounts of water drank
geom_boxplot() +
# setting nice colors for my temperature values (intuitive ones)
scale_color_manual(values = c("Cold" = "skyblue", # blue for cold
                              "Warm" = "palevioletred", # this color is
                              ↪ just warm to me idk
                              "Hot" = "red3")) + # red for hot, duh
# renaming axes to be neater and more clear
labs(
  x = "Temperature: hot, warm, or cold", # x axis looking at temperature
  y = "Water intake (L)", # y axis looking at water intake
  # title that says something about the figure
  title = "On average, I drink more water when it is hot outside",
  # subtitle stating when the most recent data collection was
  subtitle = "Most recent data collected 5/25/2026"
) +
# my favorite theme
theme_bw() +
# removing the legend, not super necessary here
theme(legend.position = "none",
      # removing grid panels
      panel.grid = element_blank())

```

On average, I drink more water when it is hot outside

Most recent data collected 5/25/2026

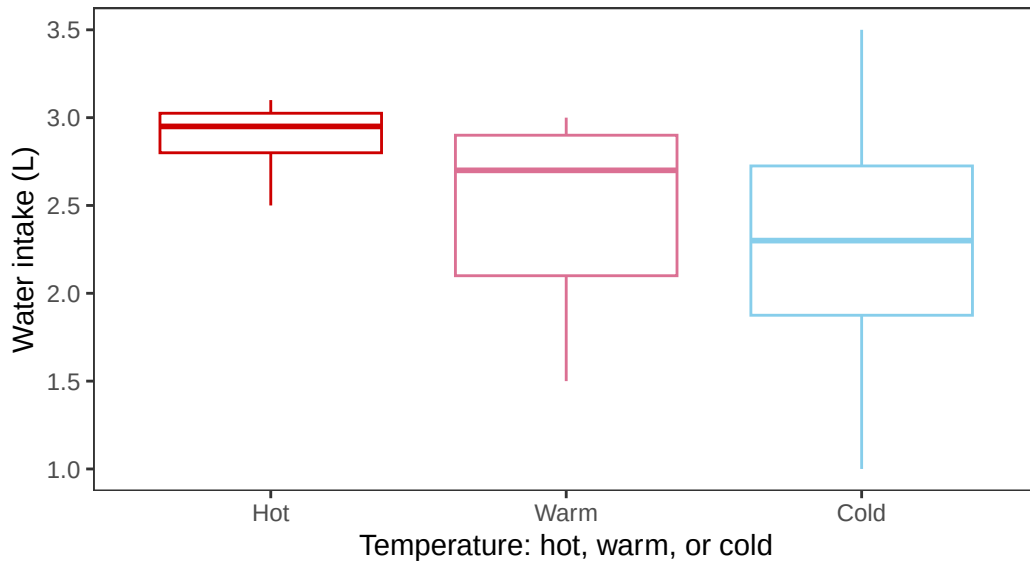


Figure 2:

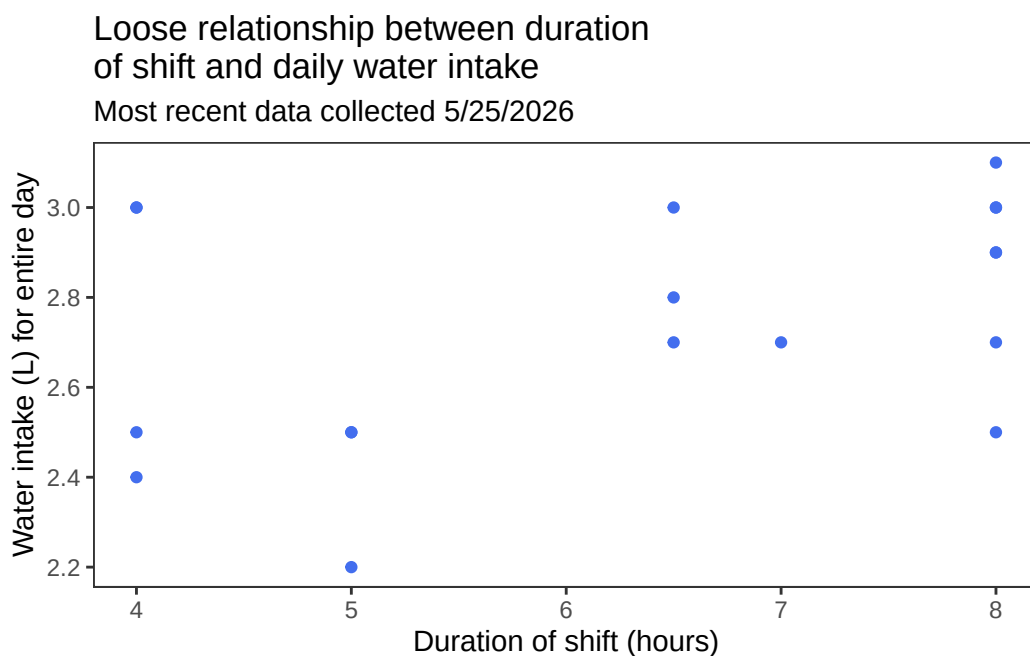
```
# making new object specifically for this part, as i do not want to mess with
↳ my other cleaned data object in case i want to re-run
my_data_steps <- personal_data |>
  # cleaning names to be lowercase and include underscores
  janitor::clean_names() |>
  # dropping n/a values for duration of shift to filter for only days i
  ↳ worked
  drop_na(duration_of_shift_hrs)
```

```
# base layer ggplot with my_data_steps data frame
ggplot(data = my_data_steps,
  # setting x and y axes
  mapping = aes(x = duration_of_shift_hrs, # x axis is duration of shift
    y = water_intake_l)) + # y axis is water intake
  # first layer geom_point in pretty royal blue color
  geom_point(color = "royalblue2") +
  # setting axes, title, and subtitles
  labs(
    # x axis title duration of shift
    x = "Duration of shift (hours)",
    # y axis title water intake
```

```

y = "Water intake (L) for entire day",
# bland statement for title because this relationship is kind of a
  ↳ stretch
title = "Loose relationship between duration \nof shift and daily water
  ↳ intake",
# subtitle most recent data collected on may 25
subtitle = "Most recent data collected 5/25/2026"
) +
# voted best theme on a reddit post i read
theme_bw() +
# getting rid of panel grid
theme(panel.grid = element_blank())

```



b. Creating captions for figures

Figure 1: Relationship between daily water intake (L) and daily temperature (hot, warm, cold)

On average, my median daily water intake (measured in liters) is higher when I experience warmer temperatures that day, as shown by the median bar for the data in red (hot) being higher than the median bar for the data in pink (warm) and the median bar for the data in blue (cold).

Figure 2: Relationship between duration of shift (hrs) and daily water intake (L)

There is not a strong correlation between the duration of the shift worked that day (measured in hours) and daily water intake (measured in liters). Based on the orientation of the blue points, which represent individual, daily observations, there may be a positive relationship between these two variables, but a large amount of outliers make it difficult to discern a concrete relationship.

Problem 3: Affective visualization

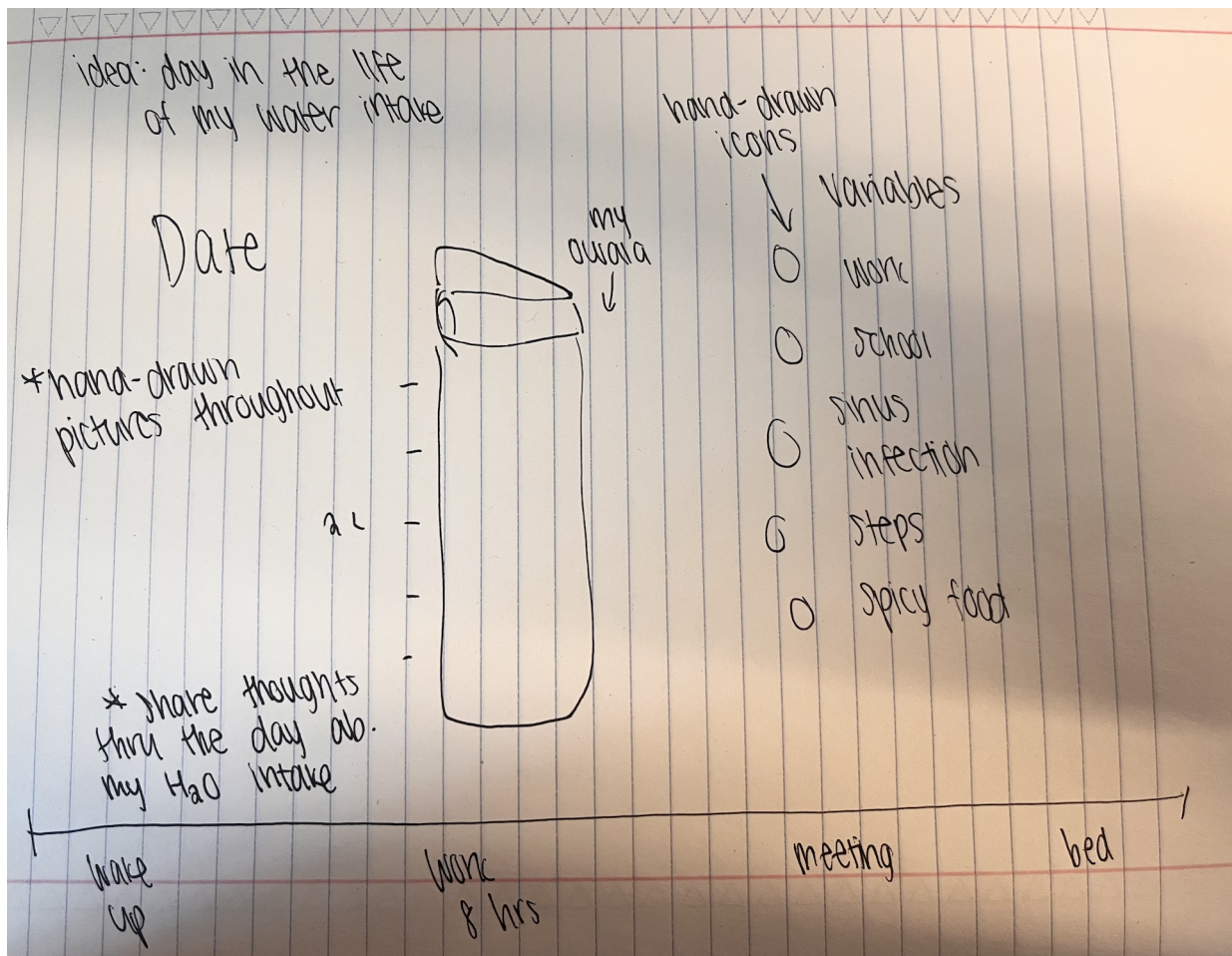
a. Describing my affective visualization

For my affective visualization, I want to utilize the fact that I am recording a ton of variables every day, including whether or not I went to school, work, if I had a sinus infection, and much more. Consequently, for my affective visualization, I would like to use my water consumption for a day to tell a story about my entire day, like a “day-in-my-life” premise. I want it to have character and reflect my personal artistic style because it is *about me*. I am thinking of drawing my Owala water bottle as the main component and visually show my water consumption, then surround this with different variable explanations about what I did that day that could have contributed to my water consumption.

b. Creating a sketch on paper of my idea

Paper sketch:

```
# code to include screenshot of affective visualization paper sketch
knitr::include_graphics(here("drawings_and_rough_visualizations",
  ↪  "stats_paper_drawing.png"))
```



Note: Sorry, my lighting is not the best!

c. Making a draft of my visualization

Rough draft:

```
# code to include screenshot of affective visualization rough draft
knitr::include_graphics(here("drawings_and_rough_visualizations",
  ↳ "stats_visualization.png"))
```

Ella's

daily
H₂O
intake

May 15

Today's temp:

COLD + GLOOMY

Today's variables:



work: yes



school: yes



sinus infection:
yes (always)



steps:
12,071



spicy food:
2x



my throat
hurts!

rushing, but
staying hydrated!

my throat
hurts...
worse?

CHUG!

REFILL!

REFILL!

CHUG!

Woke up
sick

8 hr
housekeeping
shift



research
meeting



Go to
bed...
sick

d. Beginning to form an artist statement

- For this piece, I am showing my daily water intake (L), and what conditions I experienced that day that may influence it.
- After looking over the resources provided to us, I really enjoyed the character in Stefanie Posavec and Giorgia Lupi's Dear Data project. I feel as though, in this project, they showed a lot of character, which inspired me to include many personal aspects in my work, such as my thoughts throughout the day, funny little remarks, and other small components.
- I decided to do a digital drawing because I felt as though hand-drawing my icons the way that I experience and "feel" my variables would be the most authentic. For example, my hand-drawn sad "sinus infection" face is exactly how I feel when I have a sinus infection.
- I created my work by first thinking about what I wanted to accomplish, which was telling a story about my day and mood through my work. So, I started with a time-line to outline my day and what my thoughts were about water. I added my variables as well, and then included lots of little icons that I associate with those variables.

e. Hyperlink to slideshow for 5/29 workshop

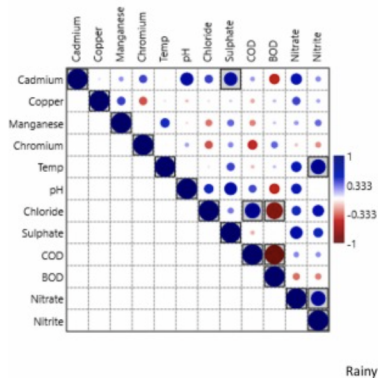
[Click here](#) to view slideshow presentation

Problem 4: Statistical critique

a. Revisiting and summarizing

For my statistical critique, I am examining a paper entitled, "Behavioural dynamics of water quality parameters and human exposure assessment of trace metals in Agulu Lake Water, South-East Nigeria" from the ScienceDirect journal. In this paper, they include a Spearman correlation, as they are focusing on examining relationships between pairs of variables. In this scenario, the response variable could be one water quality parameter (like concentration of a specific trace metal) and the predictor variable could be another water quality parameter (like another trace metal).

```
# code to include png of the spearman correlation figure from paper
knitr::include_graphics(here("data", "spearman_screenshot copy.png"))
```

b. Assessing visual clarity

I do think that the authors clearly represented their statistics in figures. Because they had so many trace metals (and relationships between these trace metals) to analyze, I believe this figure is an efficient way to visually demonstrate the correlation between two trace metals. This figure does not show explicit values per circle (Spearman correlation values), but the relative size of the circle compared to other sizes makes it easy to tell whether or not the presence of two trace metals are heavily related. The addition of the scale on the side also makes it very clear.

c. Assessing aesthetic clarity

This figure is mostly ink, as they data is represented by purposefully sized dots. Given how many relationships between trace metals this figure is exploring, I feel as though they have excellent aesthetic clarity with minimal visual clutter. This figure shows 72 (half the figure is filled because the other half would be repetitive) relationships, so I commend them on their creative figure creation to convey so much data in one figure.

d. Recommendations

The only recommendation I would have for the authors is to add a title disclosing that the figure shows the correlation between the two types of trace metals found in the water. Although it is relatively clear that is the intended purpose of the figure in the caption, I think a title would serve this figure well because it is often the first component of a figure that a reader looks at is the title, and it lets them know what to expect while analyzing the figure. It is my understanding that a great figure is at least relatively clear to readers with just the title and figure itself.